ShopSmart Retail

Data Preprocessing Report

Project: Week 2 — Data Preprocessing & Feature Engineering for Machine Learning

1. Introduction

Data preprocessing prepares raw data for analysis and machine learning by checking data quality, correcting data types, engineering useful features, examining outliers, and identifying potentially redundant or unsuitable variables.

This report documents the preprocessing performed on master_dataset.csv and the creation of shopsmart_processed.csv.

2. Dataset Overview

The original dataset contained 50,000 records and 37 columns covering orders, customers, products, stores, transaction amounts, and return status.

The original columns included identifiers such as Order_ID and Customer_ID; transaction variables such as Quantity_order, Unit_Price_order, Line_Total, and Order_Total; customer variables such as Age and Customer_Segment; product variables such as Category, Cost_Price, and Selling_Price; and the target variables Order_Total and Return_Flag.

The intended modelling tasks are:

- Regression: predict Order_Total.
- Classification: predict Return_Flag.

3. Data Inspection and Quality Checks

3.1 Initial inspection

The dataset was loaded into pandas and examined using methods including head(), shape, info(), and describe().

The initial inspection confirmed 50,000 records and 37 columns. Descriptive statistics were used to understand numerical ranges and distributions, while categorical summaries were used to examine the number of distinct values in text columns.

3.2 Missing values and duplicate records

The initial dataset contained no missing values and no duplicate rows.

After feature engineering, missing values were introduced in Customer_Tenure_Clean to represent invalid negative tenure calculations. These values were intentionally retained rather than deleting the corresponding records.

3.3 Text standardization

Leading and trailing whitespace was removed from object-type columns using pandas string operations. This helped standardize text values before further analysis.

Checks of important categorical columns showed consistent values for payment methods, sales channels, order statuses, genders, customer segments, product categories, and regions.

3.4 Date conversion

Order_Date, Registration_Date, and Opening_Date were converted to datetime data types using pd.to_datetime() with errors="coerce".

All 50,000 records retained valid values in these three date columns after conversion.

4. Feature Engineering

4.1 Order-date features

The following features were derived from Order_Date:

- Order_Year: calendar year of the order.
- Order_Month: month number of the order.
- Order_Day: day of the month.
- Order_DayOfWeek: name of the weekday.

These features make it possible to explore order patterns across different calendar periods.

4.2 Customer tenure

Customer_Tenure_Days was calculated by subtracting Registration_Date from Order_Date.

A total of 5,159 records produced negative tenure values because their order dates preceded their recorded registration dates. Rather than changing the original dates or deleting these records, a separate feature, Customer_Tenure_Clean, was created. Negative tenure values were replaced with missing values in this new feature.

The cleaned tenure feature therefore contains 5,159 missing values. These records should be investigated further, and an appropriate missing-value strategy should be selected before model training.

4.3 Validation of transaction calculations

Two consistency checks were performed:

- Line_Total matched Quantity_order × Unit_Price_order − Discount_order for all 50,000 records.
- Order_Total matched Line_Total + Shipping_Cost for all 50,000 records.

These checks confirmed consistency with the formulas represented by the dataset. They do not independently establish that every recorded amount is commercially correct.

4.4 Removal of a constant feature

Sub_Category contained only one distinct value, General, so it was removed because it could not distinguish records.

Order_Year was also removed from the final saved dataset because all records belonged to 2025, making the feature constant in this dataset.

5. Categorical Encoding

One-hot encoding was demonstrated using pd.get_dummies() for the following categorical columns:

- Customer_Segment
- Payment_Method
- Sales_Channel
- Category
- Region

The resulting df_encoded dataset contained 54 columns, compared with 42 columns before encoding. The original five categorical columns were replaced with 17 binary indicator columns.

This representation allows categorical values to be expressed numerically for algorithms that require numeric input.

Important: The one-hot-encoded dataset was a preprocessing demonstration. The final file shopsmart_processed.csv retains the original categorical columns. Model-specific encoding should be applied consistently through a training pipeline.

6. Numerical Feature Scaling

Standardization using StandardScaler was demonstrated on the following numerical features:

- Quantity_order
- Unit_Price_order
- Discount_order
- Shipping_Cost
- Age
- Cost_Price
- Selling_Price
- Order_Month
- Order_Day
- Customer_Tenure_Clean

The scaled values were checked and showed means close to zero and standard deviations close to one for the selected features.

The scaled data was retained in a separate working object, df_scaled. The final saved dataset contains the original, unscaled numerical values.

For actual model development, the scaler should be fitted only on the training data and then applied to validation and test data. Missing values must also be handled appropriately before scaling features that contain them.

7. Outlier Detection

Boxplots and the interquartile range (IQR) method were used to examine seven numerical features:

- Quantity_order
- Unit_Price_order
- Discount_order
- Shipping_Cost
- Age
- Cost_Price
- Selling_Price

No observations fell outside the calculated IQR bounds for these seven variables.

Consequently, no records were removed on the basis of this outlier check. This result applies only to the examined features and does not establish that every variable in the dataset is free from unusual values. The Order_Total boxplot by customer segment, for example, showed high-value observations.

8. Feature Selection and Leakage Considerations

Feature selection involved examining unique-value counts, relationships between variables, and the intended prediction targets.

8.1 Identifiers and redundant variables

Order_ID and Order_Item_ID were unique across all 50,000 records. These identifiers are generally unsuitable as direct numerical predictors without a specific justification.

The checks also identified potentially redundant location columns. In this dataset, each value of City mapped to one value of State, and each City_store mapped to one State_store. These relationships should be considered when selecting model inputs.

The constant features Sub_Category and Order_Year were removed as described earlier.

8.2 Target leakage

The relationship Order_Total = Line_Total + Shipping_Cost was verified for every record. Therefore, Line_Total and Shipping_Cost must not be used as predictors when predicting Order_Total if these values would reveal the target directly at prediction time.

Order_Total must also be excluded from the input features when predicting Return_Flag if it would not be legitimately available at the time of prediction or could encode information derived from the outcome.

Order_Status should be reviewed as well because its availability and meaning at prediction time could affect whether it is a valid input.

These decisions must be made separately for the regression and classification tasks. The final processed CSV retains these columns for analysis; it is not a final model-specific feature matrix.

9. Visualization Summary

Six visualizations were created to support the analysis:

1. **Order-total histogram:** Showed a concentration of lower-value transactions and fewer higher-value transactions.
2. **Category count plot:** Beauty had the most records (11,809), while Home had the fewest (8,610).
3. **Monthly order trend:** February had the lowest reported activity and May the highest, with fluctuations across the year.
4. **Numerical correlation heatmap:** Identified strong relationships between Line_Total and Order_Total, and between Cost_Price and Selling_Price.
5. **Order-total boxplot by customer segment:** Showed considerable overlap between Bronze, Silver, and Gold, with wide spreads and high-value outliers.

6. **Return-rate bar chart:** Home had the highest record-level return rate (5.13%) and Groceries the lowest (4.58%).

These findings describe the observed dataset and should not be interpreted as proof of causation.

10. Final Processed Dataset

The processed dataset was saved as shopsmart_processed.csv.

Final verification confirmed:

Check	Result
Records	50,000
Columns	41
Duplicate rows	0
Missing values	5,159
Column containing missing values	Customer_Tenure_Clean

The final file retains the original date, categorical, and numerical fields, along with the engineered date and tenure features. It was reloaded from CSV to verify that it had been saved successfully.

The original working dataset was preserved separately.

11. Limitations and Further Work

Before model training, the following steps remain necessary:

- Investigate negative customer-tenure records and decide how to handle missing tenure values.
- Define separate input features for the regression and classification tasks.
- Exclude target-leaking and unsuitable identifier columns.
- Split the data into training and testing sets before fitting imputers, encoders, scalers, or other transformations.
- Apply encoding and scaling consistently through a model pipeline where appropriate.
- Evaluate model performance using metrics suited to each task, paying particular attention to class imbalance in Return_Flag.

The return flag has a mean of approximately 0.04884, corresponding to about 4.884% positive records. This indicates class imbalance that should be considered when evaluating the classification model.

12. Conclusion

The preprocessing workflow inspected and standardized the dataset, converted date fields, engineered time and customer-tenure features, checked transaction formulas, demonstrated categorical encoding and numerical scaling, examined outliers, and documented feature-selection and leakage considerations.

The resulting shopsmart_processed.csv contains 50,000 records and 41 columns. It provides a prepared dataset for further analysis, but additional model-specific preprocessing and feature selection are still required before training reliable machine learning models.